XiangLu Xiao

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PERSONAL STATEMENT

AI4Science researcher building multi-agent architectures for autonomous experimentation in drug discovery. Expertise spans knowledge graph reasoning, molecular property prediction, and foundation models, now applied to designing agent collectives that plan, execute, and learn from experimental workflows. Currently pursuing a PhD at Imperial College London while translating industry experience into novel agentic frameworks for closed-loop scientific discovery. (https://www.linkedin.com/in/xianglu-xiao/)

RESEARCH EXPERIENC

PhD Research Imperial College London *Oct 2024 – Present*

My PhD research focuses on developing AI co-scientist systems to support scientific discovery and drug development. This includes AI-driven drug discovery workflows, agent-based data mining, and evolutionary agent-based approaches for molecular design, protein design and organoid experiment planning. The broader aim is to build intelligent, closed-loop AI systems that can assist researchers in hypothesis generation, candidate prioritisation, experimental design and iterative scientific decision-making.

CrystalXGPT analyses protein features, historical crystallisation outcomes and experimental conditions to recommend optimised protein sequence and crystallisation strategies to reduce experimental trial-and-error. (AstraZeneca)
EvoDrug learns from historical drug-design reasoning data extracted from the literature to improve molecular design capability in a manner similar to human medicinal chemistry experts.
TargetXGPT analyses disease/pathway context, protein sequence and structural annotations, binding-pocket evidence, and multi-model docking/affinity predictions to deconvolute compound targets and prioritise actionable proteins for downstream validation, reducing experimental trial-and-error in target discovery.
OrganoidGPT learns from organoid protocol literature and experimental evidence to recommend reproducible organoid culture strategies and reduce trial-and-error.

PERFESSIONAL EXPERIENCE

AI Partner & Senior Data Scientist MindRank.AI *May 2021 - Oct 2024*
Led the R&D of core AI drug-discovery platforms, including Molecule Pro™ for cheminformatics-driven molecular design and PharmKG™ for biomedical knowledge mining. Led the AI component of MDR-001, which has now advanced to Phase III clinical development.

Biological Data Scientist Exscientia *Oct 2020 - May 2021*

Contributed to the development of Exscientia's core AI module, Centaur Biologist™, by integrating PubMed literature, multi-omics datasets, clinical data and large-scale biomedical databases to support target identification.

Data Scientist Aladdin Healthcare *Sep 2018 - Oct 2020*
Contributed to Biomedical Knowledge Graph published at KDD. Worked on CT Image Deep Learning Platform development

EDUCATION

Imperial College	PhD in Bioengineering	*Oct 2024 - Now*
Imperial College	Msc in Bioengineering	*Oct 2017 - Sep 2018*
University of Bristol	BEng in Mechanical Engineering	*Oct 2014 – Sep 2017*

SELECTED PUBLICATIONS

1. Dong, Y., **Xiao, X.,** Zhuang, X. X., Wu, W., Wang, Z. Y., Zhang, S., ... & Lu, J. H. (2026). Deepdrugdiscovery identifies blood–brain barrier permeable autophagy enhancers for Alzheimer's disease. *Nature Biomedical Engineering*, 1-17.**Co-First author**
2. Niu, Z., **Xiao, X.**, Wu, W., Cai, Q., Jiang, Y., Jin, W., ... & Chen, H. (2024). PharmaBench: Enhancing ADMET benchmarks with large language models. *Scientific Data*, 11(1), 985. **Co-first Aurthor**
3. Li, A., **Xiao, X.,** Liu, G., Li, K., Ling, Y., Deng, S., ... & Su, H. (2026). Isoginkgetin protects against degeneration of ALS motor neurons via regulating the GSK-3β–TFEB signaling axis. *Pharmacological Research*, 108172.
4. Mitophagy activation driven by ABCA1/ABCG1-mediated cholesterol efflux mitigates Alzheimer's pathology. **Molecular Neurodegeneration. Co-First author**
5. A phase 2b trial of MDR-001, an AI- assisted oral GLP-1 receptor agonist, in Chinese adults with overweight or obesity without diabetes**. Under Review Co-First author**
6. PharmkGPT: A KG-Enhanced RAG Framework for Reliable Biomedical Question Answering in Alzheimer's and Delirium. **Under Review Co-First author**

Additional Publications

7. Ai, R., **Xiao, X.,** Deng, S., Yang, N., Xing, X., Watne, L. O., ... & Fang, E. F. (2025). Artificial intelligence in drug development for delirium and Alzheimer's disease. *Acta Pharmaceutica Sinica B*. Co-First author
8. Liu, H., Qin, Y., Niu, Z., Xu, M., Wu, J., **Xiao, X.**, Lei, J., Ran, T., & Chen, H. (2024). How Good are Current Pocket-Based 3D Generative Models?: The Benchmark Set and Evaluation of Protein Pocket-Based 3D Molecular Generative Models. *Journal of Chemical Information and Modeling*.
9. Chen, Q., Sun, H., Liu, H., Jiang, Y., Ran, T., Jin, X., **Xiao, X.**, ... & Niu, Z. (2023). An extensive benchmark study on biomedical text generation and mining with ChatGPT. *Bioinformatics*, 39(9), btad557.
10. Xie, C., Zhuang, X. X., Niu, Z., Ai, R., Lautrup, S., Zheng, S., **Xiao, X.**, ... & Fang, E. F. (2022). Amelioration of Alzheimer's disease pathology by mitophagy inducers identified via machine learning and a cross-species workflow. *Nature Biomedical Engineering*, 6(1), 76-93.
11. Zheng, S., Rao, J., Song, Y., **Xiao, X.**, Fang, F., Yang, Y., Niu, Z. (2020). PharmKG: A Dedicated Knowledge Graph Benchmark for Biomedical Data Mining. *Briefings in Bioinformatics*.